Solutions to Chapter 3 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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Conventions. A *transition graph* has an arrow $S \to S'$ when $f(S) \supseteq S'$. A closed path $S_1 \cdots S_k S_1$ gives a fixed point of f^k in S_1 (Corollary 3.18). An orbit is *chaotic* if it is bounded, not asymptotically periodic, and has positive Lyapunov exponent.

Exercise 3.1

$f_a(x) = a - x^2$.

(a) Fixed points solve $x^2 + x - a = 0$; there is exactly one iff $1 + 4a = 0$, so $a_1 = -\frac{1}{4}$.

(b) For $a < a_1$, $f_a(x) - x = -(x^2 + x - a) < 0$ for every x, and in fact $f_a(x) - x \le a - a_1 < 0$ is bounded away from 0. So every orbit decreases by at least a fixed amount each step and **diverges to** $-\infty$.

(c) For $a > a_1$ the fixed points are $x_\pm = \frac{1}{2}(-1 \pm \sqrt{1+4a})$, with $f_a'(x) = -2x$. For $x_- < -\frac{1}{2}$ we get $f' > 1$ (repelling). For $x_+ > -\frac{1}{2}$ the fixed point is attracting iff $2x_+ < 1$, i.e. $\sqrt{1+4a} < 2$, i.e. $a < \frac{3}{4}$. So $a_2 = \frac{3}{4}$.

(d) $f_a^2(x) - x = (x^2 + x - a)(x^2 - x + 1 - a)$, so period-two points solve $x^2 - x + 1 - a = 0$, which has real roots $x_{1,2}$ iff $a > \frac{3}{4}$. Here $x_1 x_2 = 1 - a$, so the multiplier is $f'(x_1)f'(x_2) = 4x_1x_2 = 4(1-a)$. It is attracting iff $|4(1-a)| < 1$, i.e. $\frac{3}{4} < a < \frac{5}{4}$. So $a_3 = \frac{5}{4}$.

(e) For $a = 2$, $f(x) = 2 - x^2$ has fixed points 1 and -2, both sources ($f' = -2,\ 4$). The interval $[-2, 2]$ is mapped onto itself ($f(0) = 2$, $f(\pm 2) = -2$), and every orbit starting outside it tends to $-\infty$. On $[-2, 2]$, f is conjugate to the logistic map $G(x) = 4x(1-x)$ (see 3.3), so the dynamics there is chaotic: periodic orbits of all periods, dense periodic points, sensitive dependence, and chaotic orbits.

Exercise 3.2

For $f(x) = 2x \pmod 1$, use $L = [0, \frac{1}{2}]$ and $R = [\frac{1}{2}, 1]$. Each is mapped onto $[0, 1]$, so the transition graph has all four arrows $L \to L$, $L \to R$, $R \to L$, $R \to R$ (the same as for the tent map). The schematic itineraries are $LL = [0, \frac{1}{4}]$, $LR = [\frac{1}{4}, \frac{1}{2}]$, $RL = [\frac{1}{2}, \frac{3}{4}]$, $RR = [\frac{3}{4}, 1]$, and in general the level-k subintervals appear in the binary order of the itinerary. **Difference:** the doubling map

is increasing on both pieces, so the order is LL, LR, RL, RR, whereas for the tent map the right branch reverses orientation and the order is LL, LR, RR, RL.

Exercise 3.3

(a) Try $C(x) = \alpha x + \beta$ with $C \circ G = g \circ C$. Then $4\alpha x(1-x) + \beta = 2 - (\alpha x + \beta)^2$. Matching coefficients: x^2: $-4\alpha = -\alpha^2$, so $\alpha = 4$; x: $4\alpha = -2\alpha\beta$, so $\beta = -2$; constant: $\beta = 2 - \beta^2$ holds. So

$$C(x) = 4x - 2, \qquad C : [0,1] \to [-2,2],$$

and indeed $g(C(x)) = 2 - (4x-2)^2 = 16x - 16x^2 - 2 = C(G(x))$.

(b) G has chaotic orbits (Theorem 3.13). Let $\{y_n\}$ be one and set $z_n = C(y_n)$, an orbit of g since $C \circ G = g \circ C$. It is bounded. It is not asymptotically periodic, since C is an affine bijection ($|z_n - C(p_n)| = 4|y_n - p_n|$ and C maps periodic orbits to periodic orbits). Finally $g'(C(y)) = -2(4y-2) = 4 - 8y = G'(y)$, so the Lyapunov exponents agree, $h_g(z_1) = h_G(y_1) > 0$. Hence z_n is a chaotic orbit of g.

Exercise 3.4

$g(x) = 2.5x(1-x)$ has fixed points 0 and $p = 0.6$, with $g'(p) = 2 - 2.5 = -0.5$, so p is a sink. A chaotic orbit must be bounded and not asymptotically periodic, so it suffices to show that every bounded orbit is asymptotically periodic.

- If $x < 0$ or $x > 1$, then (after one step if $x > 1$) the orbit is negative and decreases to $-\infty$: unbounded. If $x = 0$ or 1 the orbit is fixed at 0 after at most one step.
- If $x \in (0,1)$, then $g(x) \in (0, 0.625]$. Let $J = [0.5, 0.625]$. Then $g(J) = [0.5859, 0.625] \subset J$ and $|g'(x)| = 2.5|1-2x| \le 0.625 < 1$ on J, so every orbit in J converges to p. An orbit starting in $(0, 0.5)$ increases (g is increasing there and $g(x) > x$) and cannot converge to a fixed point in $(0, 0.5]$ (there is none), so it reaches $(0.5, 0.625) \subset J$. Hence it also converges to p.

So every bounded orbit is asymptotically periodic and there are no chaotic orbits.

Exercise 3.5

$f(x) = -2x^2 + 8x - 5$.

(a) This is a downward parabola with vertex $(2,3)$, zeros at $x = 2 \pm \frac{\sqrt{6}}{2} \approx 0.775, 3.225$, and key values $f(1) = 1$, $f(2) = 3$, $f(3) = 1$. Its fixed points are 1 and 2.5.

(b) Take $I_1 = [1,2]$ and $I_2 = [2,3]$.

(c) f is increasing on I_1 with $f(I_1) = [1,3]$, and decreasing on I_2 with $f(I_2) = [1,3]$. So each interval covers both, and the transition graph has all four arrows $I_1 \to I_1, I_2$ and $I_2 \to I_1, I_2$. For every n there is a closed path $I_1 \cdots I_1 I_2 I_1$ of length n whose symbol sequence is not a repetition of a shorter block, so f has periodic orbits of **every period** $1, 2, 3, \ldots$.

Exercise 3.6

$f(x) = 2x(1-x)$ has vertex $(\frac{1}{2}, \frac{1}{2})$, with $f(0) = f(1) = 0$; its fixed points are 0 and $\frac{1}{2}$. Take $L = [0, \frac{1}{2}]$ and $R = [\frac{1}{2}, 1]$. Then $f(L) = [0, \frac{1}{2}] = L$ (increasing) and $f(R) = [0, \frac{1}{2}] = L$ (decreasing). The transition graph has only the arrows $L \to L$ and $R \to L$. Any closed path consists only of L's, and nothing maps onto R, so the only periodic orbits are fixed points: the only possible period is **1**. (Indeed, every orbit in $(0, 1)$ converges to the superstable fixed point $\frac{1}{2}$.)

Exercise 3.7

Let $J \subset [0, 1]$ be an interval and choose n with $2^{-n} < |J|$. Then J contains a dyadic interval $D = [a, b]$ of length 2^{-n} on which T^n is linear with slope $\pm 2^n$ and maps D onto $[0, 1]$. So $\{T^n(a), T^n(b)\} = \{0, 1\}$, and $T^n(x) - x$ has opposite (weak) signs at a and b: if $T^n(a) = 0$, $T^n(b) = 1$, then $-a \leq 0$ and $1 - b \geq 0$; if $T^n(a) = 1$, $T^n(b) = 0$, then $1 - a \geq 0$ and $-b \leq 0$. By the intermediate value theorem T^n has a fixed point in $D \subset J$, which is a periodic point of T. Hence periodic points are dense.

Exercise 3.8

Let $I_i = [x_i, x_{i+1}]$ for $i = 1, \ldots, 8$.

(a) Sharkovskii's ordering is $3 \succ 5 \succ 7 \succ 9 \succ \cdots \succ 2 \cdot 3 \succ 2 \cdot 5 \succ \cdots \succ 2^3 \succ 2^2 \succ 2 \succ 1$. A period-9 orbit guarantees periods 1, every even number, and every odd number ≥ 9 $(9, 11, 13, \ldots)$. It does *not* by itself guarantee $3, 5, 7$.

(b) Since $f(x_i) = x_{i+1}$, we get $f(I_i) \supseteq I_{i+1}$ for $i = 1, \ldots, 7$. Also $f(x_8) = x_9$ and $f(x_9) = x_1$ give $f(I_8) \supseteq [x_1, x_9] = I_1 \cup \cdots \cup I_8$. The transition graph has arrows $I_i \to I_{i+1}$ $(1 \leq i \leq 7)$ and $I_8 \to I_j$ for all $j = 1, \ldots, 8$. For each $k = 1, \ldots, 8$ the closed path $I_8\, I_{9-k}\, I_{10-k} \cdots I_7\, I_8$ has length k and contains I_8 only once, so it gives a point of exact period k. For $m \geq 1$, the closed path $I_8^m\, I_1 I_2 \cdots I_7\, I_8$ has length $m + 7$ and contains I_1 once per cycle, giving every period ≥ 8. So **every period** $1, 2, 3, \ldots$ is certain to exist (in particular $3, 5, 7$, which Sharkovskii alone does not guarantee).

Exercise 3.9

Here $x_1 \to x_2 \to x_4 \to x_3 \to x_1$, so $\{x_1, x_2, x_3, x_4\}$ is a period-4 orbit.

(a) The graph passes through (x_1, x_2), (x_2, x_4), (x_3, x_1), (x_4, x_3). So f increases on $[x_1, x_2]$ (from x_2 to x_4), decreases on $[x_2, x_3]$ (from x_4 to x_1), and increases on $[x_3, x_4]$ (from x_1 to x_3): an "N-shaped" graph.

(b) Let $A = [x_1, x_2]$, $B = [x_2, x_3]$, $C = [x_3, x_4]$. Then $f(A) = [x_2, x_4] = B \cup C$, $f(B) = [x_1, x_4] = A \cup B \cup C$ and $f(C) = [x_1, x_3] = A \cup B$. The arrows are $A \to B, C$; $B \to A, B, C$; $C \to A, B$.

(c) Closed paths: B (period 1), BAB (period 2), $BACB$ (period 3), and $B^{n-2}AC\,B$ for each $n \geq 3$ (length n, with A appearing once). So **every period** $1, 2, 3, \ldots$ must exist.

Exercise 3.10

Here $x_1 \to x_3 \to x_4 \to x_2 \to x_5 \to x_1$ is a period-5 orbit.

(a) The graph passes through (x_1, x_3), (x_2, x_5), (x_3, x_4), (x_4, x_2), (x_5, x_1): it rises from x_3 to x_5 on $[x_1, x_2]$, falls from x_5 to x_4 on $[x_2, x_3]$, falls from x_4 to x_2 on $[x_3, x_4]$, and falls from x_2 to x_1 on $[x_4, x_5]$.

(b) With $I_i = [x_i, x_{i+1}]$ $(i = 1, \dots, 4)$: $f(I_1) = [x_3, x_5] = I_3 \cup I_4$, $f(I_2) = [x_4, x_5] = I_4$, $f(I_3) = [x_2, x_4] = I_2 \cup I_3$, $f(I_4) = [x_1, x_2] = I_1$. The arrows are $I_1 \to I_3, I_4$; $I_2 \to I_4$; $I_3 \to I_2, I_3$; $I_4 \to I_1$.

(c) Closed paths: I_3 (period 1); $I_4 I_1 I_4$ (period 2); and $I_3^{n-3} I_2 I_4 I_1 I_3$ for each $n \geq 4$ (length n, with I_2 appearing once). So periods $1, 2$ and every period ≥ 4 must exist. The graph has no closed path of length 3 that is not a repetition of a fixed point, so period 3 is *not* guaranteed; this is consistent with Sharkovskii's theorem for a period-5 orbit.

Exercise 3.11

(a) Let $h(x) = f(x) - x$. Then $h(0) = f(0) \geq 0$ and $h(1) = f(1) - 1 \leq 0$, so h has a zero in $[0, 1]$.

(b) Since f is onto, there are a, b with $f(a) = 0$ and $f(b) = 1$.

Claim: f or f^2 has points $x_1 < x_2$ with $x_1 \mapsto 0$ and $x_2 \mapsto 1$. If some choice has $a < b$, take $x_1 = a$, $x_2 = b$ for f. Otherwise $b < a$ for every such choice. Then $f([b, a]) \supseteq [0, 1]$, because $f(b) = 1$ and $f(a) = 0$. Let $y = \min\{t \in [b, a] : f(t) = a\}$ (this exists since $f(b) = 1 \geq a \geq f(a) = 0$) and let $x = \min\{t \in [y, a] : f(t) = b\}$ (this exists since $f(y) = a > b$ and $f(a) = 0 \leq b$). Then $y < x$ (as $f(y) = a \neq b = f(x)$), $f^2(y) = f(a) = 0$ and $f^2(x) = f(b) = 1$.

Now let h be f or f^2 with $h(x_1) = 0$, $h(x_2) = 1$, $x_1 < x_2$. The sign of $h(x) - x$ is ≥ 0 at 0, ≤ 0 at x_1 (value $-x_1$), ≥ 0 at x_2 (value $1 - x_2$), and ≤ 0 at 1. So h has a fixed point in each of $[0, x_1]$, $[x_1, x_2]$, $[x_2, 1]$. These are distinct unless $x_1 = 0$ or $x_2 = 1$, and in those cases h has the fixed point 0 (resp. 1) together with a fixed point in $[x_2, 1]$ (resp. $[0, x_1]$), which is another point. So h has at least two fixed points, and these are fixed points of f^2 (as $f^2 = h$ or $f^2 = h^2$).

(c) If $0, 1$ are not fixed by f, then $f(0) > 0$ and $f(1) < 1$, so f has a fixed point $p \in (0, 1)$ by (a). Take h, $x_1 < x_2$ as in (b). If $h = f$, then $a = x_1 > 0$ (as $f(0) > 0$) and $b = x_2 < 1$ (as $f(1) < 1$), and the sign changes are strict at x_1, x_2, so f has three distinct fixed points in $[0, x_1)$, (x_1, x_2), $(x_2, 1]$, hence so does f^2. If $h = f^2$ and $0 < x_1$, $x_2 < 1$, the same argument gives three distinct fixed points of f^2. If $x_1 = 0$, then $f^2(0) = 0$ with $f(0) \neq 0$, so $\{0, f(0)\}$ is a period-two orbit, and together with p this gives three fixed points of f^2 ($f(0) \neq p$, since otherwise $f^2(0) = p \neq 0$). If $x_2 = 1$, then $\{1, f(1)\}$ is a period-two orbit, and we again get three fixed points of f^2 with p.

Exercise 3.12

$f(x) = rx(1-x)$, $f'(x) = r(1-2x)$, $r > 2+\sqrt{5}$ (so $r^2 - 4r > 1$). Write $u = x - \frac{1}{2}$. Then $f(x) = r(\frac{1}{4} - u^2) > 1$ iff $|u| < \delta_0 := \frac{1}{2}\sqrt{1-4/r}$. Any orbit that stays in $[0,1]$ must therefore satisfy $|x_i - \frac{1}{2}| \geq \delta_0$ for all i (otherwise $x_{i+1} > 1$), so

$$|f'(x_i)| = 2r|x_i - \tfrac{1}{2}| \geq 2r\delta_0 = \sqrt{r^2 - 4r} =: \lambda > 1.$$

Hence $\frac{1}{n}\sum_{i=1}^{n} \ln|f'(x_i)| \geq \ln\lambda > 0$ for all n, and if the limit defining the Lyapunov exponent exists then $h \geq \ln\lambda > 0$.

Exercise 3.13

Take $n \geq 2$ (for $n = 1$ every point is fixed). Then $f^k(x) = n^k x \pmod 1$.

- **Periodic:** $f^k(x) = x$ iff $(n^k - 1)x \in \mathbb{Z}$, i.e. $x = j/(n^k - 1)$. These are exactly the rationals p/q in lowest terms with $\gcd(q, n) = 1$ (for such q, $n^k \equiv 1 \pmod q$ for some k).
- **Eventually periodic:** exactly the rationals $x \in [0,1]$. For $x = p/q$, split $q = q_1 q_2$ with q_1 built from primes dividing n and $\gcd(q_2, n) = 1$; after enough steps the orbit lands on a periodic point. Irrationals are never eventually periodic.
- **Asymptotically periodic:** exactly the eventually periodic points. Suppose $|x_k - y_k| \to 0$ for a periodic orbit y_k. The distance $d_k = x_k - y_k \pmod 1$ satisfies $d_{k+1} = nd_k \pmod 1$, so once $0 < |d_k| < \frac{1}{2n}$ it is multiplied by $n \geq 2$ at each step, and it can never tend to 0 unless it equals 0 from some point on.
- **Chaotic orbits:** the orbit of x is bounded, and wherever it is defined $|f'| = n$, so the Lyapunov exponent is $\ln n > 0$ for orbits avoiding the discontinuities k/n (all irrational orbits do). So the chaotic orbits are exactly the orbits of the **irrational** points, those that are not asymptotically periodic.

Exercise 3.14

We prove the theorem by contradiction (this also proves fact (ii)). Suppose some orbit $\{x_n\}$ is dense in I, f has two distinct fixed points, and f has no point of period 3.

Preliminaries. Let F be the set of fixed points. (1) The orbit never lands on F: otherwise it is eventually constant and its closure is finite, not I. (2) Any tail $\{x_n\}_{n \geq N}$ is still dense (removing finitely many points from a dense subset of an interval leaves a dense set), so every tail visits every nonempty open subset of I.

Setup. The orbit visits the open interval between the two given fixed points at some point $x \notin F$. Let $c_1 = \sup\{c \in F : c \leq x\} < x < c_2 = \inf\{c \in F : c \geq x\}$; these are fixed points (F is closed) with no fixed point in (c_1, c_2). By the intermediate value theorem $f(t) - t$ has constant sign on (c_1, c_2); replacing f and x by $-f(-x)$ and $-x$ if necessary (this preserves all hypotheses) we may assume $f(t) > t$ on (c_1, c_2).

Step 1: $c_1 = \min I$. Let $x_{n_0} \in (c_1, c_2)$. Apply fact (i) with $a = x_{n_0}$ and $b = c_1$: if $f^k(a) \le c_1$ for some $k > 1$ there would be a period-3 point. So $x_n > c_1$ for all $n \ge n_0$. If I had points below c_1, the tail would visit them; contradiction.

Step 2: orbit points $a > c_2$ satisfy $f(a) < a$. If $f(a) > a$ for such $a = x_m$, apply (i) with $b = c_2$. The tail visits (c_1, c_2), so $f^k(a) < c_2 = b$ for some $k > 1$, which gives period 3. Since $a \notin F$, $f(a) < a$.

Step 3: the orbit tail lies in $[c_1, z]$, where $z = \max_{[c_1,c_2]} f$. A tail point $x_n > c_2$ comes from an earlier point of (c_1, c_2) followed by a chain of points $> c_2$; the first of these is $\le z$ and, by Step 2, they decrease.

Step 4: a point mapping to c_1. For small $\varepsilon > 0$ let $n_1 > n_0$ be the first time with $x_{n_1} \in (c_1, c_1 + \varepsilon)$ (taking $\varepsilon < x_{n_0} - c_1$). Then $p_\varepsilon = x_{n_1 - 1} \ge c_1 + \varepsilon$ and $f(p_\varepsilon) < c_1 + \varepsilon$. If $p_\varepsilon < c_2$ we would have $f(p_\varepsilon) > p_\varepsilon \ge c_1 + \varepsilon$, so $p_\varepsilon > c_2$, and $p_\varepsilon \le z$ by Step 3. Letting $\varepsilon \to 0$ and passing to a convergent subsequence, we get $p^* \in [c_2, z]$ with $f(p^*) = c_1$, and $p^* \ne c_2$ since $f(c_2) = c_2$.

Step 5: period 3. By the intermediate value theorem $f([c_1, c_2]) \supseteq [c_1, z] \ni p^*$, so there is $a \in (c_1, c_2)$ with $f(a) = p^*$. Then $a < f(a)$, $b = c_1 < a$ with $f(b) \le b$, and $f^2(a) = f(p^*) = c_1 \le b$ with $k = 2 > 1$. By fact (i) f has a point of period 3, a contradiction.

Hence if some orbit is dense in I and f has two distinct fixed points, f has a point of period 3. ■

Exercise 3.15

(a) The shuffle is $f(n) = 2n - 51\varepsilon$ with $\varepsilon \in \{0, 1\}$, so by induction $f^k(n) = 2^k n - 51 m_k$ for an integer m_k. For $k = 8$, $f^8(n) = 256n - 51m \equiv n \pmod{51}$, because $255 = 5 \cdot 51$. Cards 0 and 51 are fixed. For $1 \le n \le 50$, $f(n) \in \{1, \ldots, 50\}$, so $f^8(n)$ also lies in $\{1, \ldots, 50\}$; two numbers in this range congruent mod 51 are equal. So $f^8(n) = n$ for every n: eight perfect shuffles restore the deck.

(b) The map is $f(x) = 2x \pmod{51}$, i.e. the doubling map on $[0, 51]$ rescaled by 51. Points of period dividing k are $x = 51j/(2^k - 1)$. The point $x = 51/(2^k - 1)$ has exact period k, since $2^m \not\equiv 1 \pmod{2^k - 1}$ for $0 < m < k$. So **every positive integer** occurs as a period (the integer points have periods dividing 8: $1, 2, 4, 8$).

Exercise 3.16

$f(x) = 1 - 3x$ on $[0, \frac{1}{3}]$ and $f(x) = x - \frac{1}{3}$ on $[\frac{1}{3}, 1]$; $I = [0, \frac{1}{3}]$, $J = [\frac{1}{3}, \frac{2}{3}]$, $K = [\frac{2}{3}, 1]$. Here $0 \to 1 \to \frac{2}{3} \to \frac{1}{3} \to 0$ is the period-4 orbit.

(a) $f(I) = [0, 1] = I \cup J \cup K$, $f(J) = [0, \frac{1}{3}] = I$, $f(K) = [\frac{1}{3}, \frac{2}{3}] = J$. The transition graph has arrows $I \to I, J, K$; $J \to I$; $K \to J$.

(b) In this graph every J is followed by I and every K by JI, so closed paths are cyclic concatenations of the blocks I, JI, KJI. On a given itinerary f^n is affine with slope $(-3)^m \ne 1$ ($m \ge 1$ is the number of I's), so each primitive cyclic word gives exactly one periodic orbit, found by solving the affine equations. Below, "cycle" lists where each point goes, starting from a_1.

Period	Points $a_1 < \cdots < a_p$	Cycle	Itinerary of a_1
1	$\frac{1}{4}$	$a_1 \to a_1$	I
2	$\frac{1}{6}$, $\frac{1}{2}$	$a_1 \to a_2 \to a_1$	IJ
3	$\frac{1}{8}$, $\frac{7}{24}$, $\frac{5}{8}$	$a_1 \to a_3 \to a_2 \to a_1$	IJI
3	$\frac{1}{12}$, $\frac{5}{12}$, $\frac{3}{4}$	$a_1 \to a_3 \to a_2 \to a_1$	IKJ
4	0, $\frac{1}{3}$, $\frac{2}{3}$, 1	$a_1 \to a_4 \to a_3 \to a_2 \to a_1$	$IKJI$
4	$\frac{1}{7}$, $\frac{5}{21}$, $\frac{2}{7}$, $\frac{4}{7}$	$a_1 \to a_4 \to a_2 \to a_3 \to a_1$	$IJII$
5	$\frac{11}{80}$, $\frac{19}{80}$, $\frac{61}{240}$, $\frac{23}{80}$, $\frac{47}{80}$	$a_1 \to a_5 \to a_3 \to a_2 \to a_4 \to a_1$	$IJIII$
5	$\frac{1}{28}$, $\frac{19}{84}$, $\frac{9}{28}$, $\frac{47}{84}$, $\frac{25}{28}$	$a_1 \to a_5 \to a_4 \to a_2 \to a_3 \to a_1$	$IKJII$
5	$\frac{11}{84}$, $\frac{5}{28}$, $\frac{23}{84}$, $\frac{13}{28}$, $\frac{17}{28}$	$a_1 \to a_5 \to a_3 \to a_2 \to a_4 \to a_1$	$IJIIJ$
5	$\frac{1}{24}$, $\frac{5}{24}$, $\frac{3}{8}$, $\frac{13}{24}$, $\frac{7}{8}$	$a_1 \to a_5 \to a_4 \to a_2 \to a_3 \to a_1$	$IKJIJ$

(For the orbit through 0, the endpoints $\frac{2}{3}, \frac{1}{3}$ are assigned to J and I respectively, to follow the allowed arrows.) As a check, the number of closed paths of length n is $\operatorname{tr} M^n = 1, 3, 7, 11, 21$ for $n = 1, \ldots, 5$, which matches the count of cyclic rotations of the words above.

(c) An orbit with itinerary $IJIJ$ satisfies $x_1 \in I$, $x_2 = 1 - 3x_1 \in J$, $x_3 = x_2 - \frac{1}{3} \in I$, $x_4 = 1 - 3x_3 \in J$ and $x_1 = x_4 - \frac{1}{3}$. This gives $x_1 = 9x_1 - \frac{4}{3}$, so $x_1 = \frac{1}{6}$, the period-two point $\{\frac{1}{6}, \frac{1}{2}\}$ (itinerary IJ repeated). This is the only point with that itinerary, so there is **no** period-4 orbit with itinerary $IJIJ$.